

Reward Components

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This table summarizes the reward logic used in the staged RL pipeline. Values below describe the raw reward behavior before phase-specific weighting.

Reward Type	What It Encourages	When It Gives High Reward	Simple Example
Correctness	Get the right answer	Exact normalized match with the ground truth, or the correct option letter for multiple-choice.	Gold = 42, model = 42.0 → high reward
Formatting	Follow the required structure	Exactly one REASONING block and one SOLUTION block; malformed answers reduce the score.	Clean reasoning + solution blocks → high reward
Parseability	Make the solution machine-readable	The solution can be parsed as a valid number or valid option letter.	“42” → parseable; “forty-two” → low reward
Finished	End the response cleanly	The answer is complete and stays below the token limit.	Properly closed answer → high reward; truncated at cap → low reward
Tolerance	Be approximately correct on numeric tasks	The answer is numerically close enough, but not already exact.	Gold = 100, model = 99.999 → high reward
Brevity	Stay concise	The response is short relative to the allowed completion budget.	20 tokens → high reward; very long rambling answer → low reward

Note. The required tags in code are exactly `<REASONING>` and `<SOLUTION>`. Tolerance reward is enabled only in later phases and does not add extra reward when the answer is already exact.